RAJALAKSHMI ENGINEERING COLLEGE

Laxmi Gayathiri S 251801157-2025-AIDS

L2

Started on	Monday, 24 August 2026, 3:23 PM
State	Finished
Completed on	Thursday, 8 October 2026, 11:11 AM
Time taken	44 days 19 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
    {
         i++;
         s += i;
     }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

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Falling back to raw text area.

```
    int n;
    c++;
    scanf("%d",&n);
    int i=1;
    c++;
    int s=1;
    c++;
    while(s<=n){
        c++;
        i++;
        c++;
        s+=i;
        c++;
    }
    printf("%d",c);
    return 0;

}
```

	Input	Expected	Got	
✔	9	12	12	✔
✔	4	9	9	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Laxmi Gayathiri S 251801157-2025-AIDS L2

Started on	Thursday, 6 August 2026, 10:38 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:11 AM
Time taken	63 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
      printf("*");
    }
    else
    {
     for(int i=1; i<=n; i++)
     {
       for(int j=1; j<=n; j++)
       {
          printf("*");
          printf("*");
          break;
       }
     }
   }
 }
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
#include<stdio.h>
int main(){
    int c=0;
    int n;
    c++;
    scanf("%d",&n);


    if(n==1)
    {
        c++;
        c++;

    }
    else
    {
        c++;
     for(int i=1; i<=n; i++)

     {
         c++;
       for(int j=1; j<=n; j++)
       {
           c++;
           c++;
           c++;
           c++;


          break;
       }
     }
   }
   printf("%d",c);
   return 0;
```

```
36
37
38 }
```

	Input	Expected	Got	
✔	2	12	12	✔
✔	1000	5002	5002	✔
✔	143	717	717	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Laxmi Gayathiri S 251801157-2025-AIDS L2

Started on	Thursday, 6 August 2026, 11:10 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:12 AM
Time taken	63 days
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
   for (i = 1; i <= num;++i)
   {
    if (num % i== 0)
       {
         printf("%d ", i);
       }
    }
 }
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
int main()
{
    int num;
    scanf("%d",&num);
    int count=0;
    count++;

   void Factor(int num)
    {


        for(int i=1;i<=num;++i)
        {
            count++;
            if(num%i==0)
            {

                count++;
            }
            count++;
        }
    }
    Factor(num);
    printf("%d",count);
    return 0;
}
```

	Input	Expected	Got	
✔	12	31	31	✔
✔	25	54	54	✔
✔	4	12	12	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Laxmi Gayathiri S 251801157-2025-AIDS L2

Started on	Thursday, 6 August 2026, 11:32 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:12 AM
Time taken	62 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include<stdio.h>
void function(int n)
{
    int c=0,count=2;
    for(int i=n/2;i<n;i++){
        count=count+2;
        for(int j=1;j<n;j=2*j){
            count=count+2;
            for(int k=1;k<n;k=k*2){
                count=count+1;
                c++;
                count=count+1;
            }
        }
    }
    printf("%d",count);
}
int main()
{
    int n;
    scanf("%d",&n);
    function(n);
}
```

	Input	Expected	Got	
✔	4	30	30	✔
✔	10	212	212	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Laxmi Gayathiri S 251801157-2025-AIDS

L2

Started on	Thursday, 6 August 2026, 11:43 AM
State	Finished
Completed on	Thursday, 8 October 2026, 11:12 AM
Time taken	62 days 23 hours
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```c
#include <stdio.h>

int main() {
    int n;
    // Read the positive integer n
    if (scanf("%d", &n) != 1) return 0;

    int count = 0;

    // Initialization: rev = 0 (1 step)
    int rev = 0;
    count++;

    int remainder;

    // The while loop condition runs once for every iteration,
    // plus one final time when it evaluates to false (n == 0).
    while (n != 0) {
        count++; // For the while (n != 0) true condition

        remainder = n % 10;
        count++; // For remainder assignment & modulo arithmetic

        rev = rev * 10 + remainder;
        count++; // For rev assignment, multiplication & addition

        n /= 10;
        count++; // For n division & assignment
    }
    count++; // For the final while (n != 0) false condition

    // The printf(rev) from the algorithm is part of the logic execution
    count++; // For printing the reversed value (conceptually executed)

    // Print the final accumulated counter value as requested
    printf("%d\n", count);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔
✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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